

Leggett–Garg K_3 Values Above 1 in Fibonacci–Anyon Braiding: 99.998% of the Lüders Bound, and Exactly 1 for Ising Braiding

Berkay Yüksel Sayim^{1,*}

¹*Independent Research, Germany*

ORCID: 0009-0004-4993-7352

(Dated: 2026-10-03)

We numerically test the three-time Leggett–Garg inequality $K_3 \leq 1$ for the standard B_3 Fibonacci–anyon braiding representation on the two-dimensional fusion space of three τ anyons. Exhaustive enumeration over all 4^L braid words up to length $L = 11$ and random sampling to $L = 40$ show that K_3 saturates the Lüders bound $3/2$ to 99.998%, with the first violation already at $L = 3$. Two structural signatures accompany the saturation. First, replacing the Fibonacci generators by the Ising–anyon generators on the same 2D fusion space gives $K_3 = 1$ exactly for every $L \leq 11$ in the exhaustive search and every random word tested at even $L \in \{12, 14, \dots, 40\}$ —a sharp split that mirrors the Howard–Vala no-Bell-violation result for Ising braiding in the spatial CHSH setting. Second, an algebraic phase-deformation parameter δ tunes a singular point $\delta = 3\pi/5$ at which the generator σ_1 collapses to a scalar to machine precision, so that every braid word reduces to a global phase. The Fourier-period analysis of the δ dependence reported in earlier versions of this letter is withdrawn. As a consistency check, we confirm that K_3 for the optimal $L = 11$ word is initial-state independent (every pure state and the maximally mixed state agree to $\sim 10^{-15}$, machine precision), as required by a generic $d = 2$ trace identity for qubit observables. All Yang–Baxter, unitarity, and $(\sigma_1\sigma_2)^3$ –scalar sanity checks pass at machine precision for the undeformed generators ($\delta = 0$). The closest previously published work we identified, “Leggett–Garg on a topological system,” is Gómez–Ruiz *et al.* (2018), which differs in three ways: the system is abelian (Kitaev chain, not Fibonacci); the qubit basis is formed by paired edge Majorana modes rather than the fusion channel of three anyons; and K_3 is used as a probe of a topological phase transition rather than as a saturation test. Code, seeds, and data are released with the preprint.

I. INTRODUCTION

The Bell inequalities probe whether spatially separated subsystems can be described by a joint local-realistic distribution; the Leggett–Garg inequalities [1] probe the temporal analogue—whether a single system carries a definite value of an observable between measurements, independent of whether it is measured (“macrorealism”), and under what conditions quantum mechanics can violate it [6]. For three projective measurements of a dichotomic observable $Q \in \{+1, -1\}$ at times $t_1 < t_2 < t_3$, with two-time correlators $C_{ij} \equiv \langle Q(t_i) Q(t_j) \rangle$, the simplest combination

$$K_3 = C_{12} + C_{23} - C_{13} \quad (1)$$

satisfies $K_3 \leq 1$ under macrorealism, $K_3 \leq 3/2$ under Lüders projective quantum measurement on a two-level system (the “temporal Tsirelson bound” [2]), and $K_3 \leq 3$ algebraically.

Non-Abelian anyons are the natural laboratory for asking whether a *topological* two-level system—a fusion-channel qubit whose algebra has no tensor-product factorization into spatial subsystems [7]—also saturates the Lüders bound. The Fibonacci anyon model is the smallest model in which braiding alone is computationally universal (dense in $SU(2)$ on the relevant fusion space),

while the Ising model, including its Majorana-zero-mode realizations, gives only the Clifford group and is provably unable to violate the spatial CHSH inequality when only topologically protected stabilizer operations are performed [10]; Clarke, Sau, and Das Sarma supply a non-Clifford phase gate for Majorana wires [9]. The two models therefore form a natural *universality split*: a property that requires arbitrarily close approximation of generic unitaries should discriminate sharply between them.

The temporal counterpart has not, to our knowledge, been measured. The closest published work we identified, “Leggett–Garg on a topological system,” is Gómez–Ruiz *et al.* [12], who tested K_3 for Dirac qubits formed from edge Majorana modes of the abelian Kitaev chain. We state up front how the present work differs, because this demarcation defines the novelty claim:

	Gómez–Ruiz 2018 [12]	this work
anyon type	abelian	non-Abelian
qubit basis	edge Majoranas	3τ channel
braiding	none; platform Clifford-only	universal
role of K_3	transition probe	saturation test
saturation	none reported	99.998% of $3/2$

The two papers are methodologically and physically distinct sub-cases of “Leggett–Garg on a topological system”. Apart from Brennen *et al.* [11], who construct explicit CHSH-violating settings on the Fibonacci/Ising fusion spaces but do not test the temporal inequality, we identified no further Bell or Leggett–Garg work for Fibonacci or Ising anyons. Recent solid-state and quantum

* berkasa@tutamail.com

processor demonstrations of non-Abelian topological order [14–17] provide a plausible target for an experimental test but have, to our knowledge, not yet been used for one.

We present the simplest possible numerical setup for the temporal question. Section II fixes the generators, correlator, and sanity checks. Section III reports three connected findings: (i) Fibonacci Lüders saturation to 99.998%; (ii) an exact $K_3 = 1$ for Ising at every length tested; (iii) an algebraic scalar-singularity point $\delta = 3\pi/5$. The subsection on the δ dependence (Sec. IIID) withdraws the Fourier-period analysis reported in earlier versions, and a consistency check closes Section III: the optimal K_3 does not depend on the initial state. Section IV relates the universality split to the known CHSH split and lists what the results do *not* say.

II. SETUP

A. Fusion space and observable

We work on the two-dimensional fusion space of three τ anyons with total charge τ in the Fibonacci modular tensor category [7], identified with the standard B_3 Burau-Jones representation. The dichotomic observable $Q = \sigma_z$ corresponds to a measurement of the topological charge of an anyon pair [8]; experimentally this is the interferometric fusion outcome of the pair.

B. Generators

We use the standard Fibonacci F - and R -symbols [7]:

$$F = \begin{pmatrix} 1/\varphi & 1/\sqrt{\varphi} \\ 1/\sqrt{\varphi} & -1/\varphi \end{pmatrix}, \quad R_1 = e^{-4\pi i/5}, \quad R_\tau = e^{+3\pi i/5}, \quad (2)$$

with $\varphi = (1 + \sqrt{5})/2$, giving

$$\sigma_1 = \text{diag}(R_1, R_\tau e^{i\delta}), \quad \sigma_2 = F \sigma_1 F. \quad (3)$$

Here δ is an algebraic phase-deformation parameter: $\delta = 0$ recovers the standard Fibonacci theory. For generic $\delta \neq 0$ the deformed generators do not satisfy the braid (Yang–Baxter) relation $\sigma_1 \sigma_2 \sigma_1 = \sigma_2 \sigma_1 \sigma_2$, not even up to a global phase. In $[0, 2\pi)$ it holds only at $\delta = 0$, at $\delta = 3\pi/5$, where σ_1 and σ_2 become multiples of the identity (Sec. IIIC), and at $\delta = 6\pi/5$, where they equal the complex conjugates of the $\delta = 0$ generators up to a global phase; all three points lie on the 240-point grid used below. The braid alphabet is $\{\sigma_1, \sigma_1^{-1}, \sigma_2, \sigma_2^{-1}\}$.

The Ising data [7] on the analogous 2D fusion space are

$$F_I = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad R_1^{\sigma\sigma} = e^{-i\pi/8}, \quad R_\psi^{\sigma\sigma} = e^{+i3\pi/8}, \quad (4)$$

with σ_1, σ_2 constructed identically; this is the nonuniversal Clifford branch.

C. Correlator

For projective Lüders measurement of $Q = \sigma_z$ at every time step and any initial state, the two-time correlator separating two measurements by a unitary U is

$$C(U) = \frac{1}{2} \text{Re Tr}[Z U Z U^\dagger], \quad (5)$$

with $Z = \sigma_z$. With equal-time braids B between t_1 – t_2 and t_2 – t_3 , so $U_{13} = B^2$, we have

$$K_3 = 2C(B) - C(B^2). \quad (6)$$

The correlator (5) is invariant under any global phase of U , so we may use the full unitary generators without projecting onto $\text{SU}(2)$. This closed-form expression of the correlator in terms of the connecting unitary follows the same general treatment as the temporal-CHSH correlator of Ref. [4].

D. Sanity

Before reporting results we verify generator consistency at $\delta = 0$. All checks pass at machine precision:

	Fibonacci	Ising
$\ \sigma_i \sigma_i^\dagger - I\ $	1.5×10^{-16}	5.6×10^{-16}
$\ \sigma_1 \sigma_2 \sigma_1 - \sigma_2 \sigma_1 \sigma_2\ $	2.4×10^{-16}	2.8×10^{-16}
$(\sigma_1 \sigma_2)^3$ scalar?	yes, phase $e^{2\pi i/5}$	yes, phase $e^{-i\pi/4}$
$\ (\sigma_1 \sigma_2)^3 - \omega I\ $	2.0×10^{-16}	2.5×10^{-16}

The central scalar of B_3 for Fibonacci is the fifth root of unity $e^{2\pi i/5}$ (consistent with the conformal weight $h_\tau = 2/5$ [7]); for Ising it is $e^{-i\pi/4}$ (consistent with central charge $c = 1/2$).

E. Search

For each $L \in \{1, \dots, 11\}$ we enumerate all 4^L braid words and record the largest K_3 . For $L \in \{12, \dots, 40\}$ we draw 4×10^5 uniformly random words per L . The deformation sweep samples $\delta \in [0, 2\pi)$ at 240 points, using exhaustive $L \leq 9$ at each δ . All computations use `numpy` only; the random seed is fixed (see code release).

III. RESULTS

A. Fibonacci: K_3 up to 99.998% of the Lüders bound

Table I lists the best K_3 for every $L \leq 11$. The first violation of macrorealism occurs already at $L = 3$, with $K_3 = 1.292$. The maximum over exhaustive search is $K_3^{\max} = 1.499762$ at $L = 11$ (tied with $L = 9$), attained

TABLE I. Best K_3 versus braid-word length L for the Fibonacci and Ising representations at $\delta = 0$. Exhaustive search up to $L = 11$. The Fibonacci values approach the Lüders bound $3/2$ already at $L = 9$ (exhaustive maximum 99.984%; deeper random sampling at $L \geq 22$ reaches 99.998%, never exactly; see Sec. III); the Ising values do not violate macrorealism for any tested L .

L	# words	$K_3^{(\text{Fib})}$	$C(B)^{(\text{Fib})}$	$C(B^2)^{(\text{Fib})}$	$K_3^{(\text{Ising})}$
1	4	1.00000	1.0000	1.0000	1.00000
2	16	1.00000	1.0000	1.0000	1.00000
3	64	1.29180	0.8197	0.3475	1.00000
4	256	1.40325	0.3475	-0.7082	1.00000
5	1 024	1.29180	0.8197	0.3475	1.00000
6	4 096	1.47744	0.5704	-0.3366	1.00000
7	16 384	1.43396	0.3212	-0.7915	1.00000
8	65 536	1.47744	0.5704	-0.3366	1.00000
9	262 144	1.49976	0.4977	-0.5043	1.00000
10	1 048 576	1.49783	0.4977	-0.5024	1.00000
11	4 194 304	1.49976	0.4977	-0.5043	1.00000

by the word $\sigma_1\sigma_2^2\sigma_1^{-3}\sigma_2^2\sigma_1^{-1}\sigma_2\sigma_1$ (correcting a mistranscribed 14-letter word reported in v1.0–v1.1; the corrected 11-letter word is the one recorded by the deposited search code, `lgi_fibonacci.py`, and independently re-verified by two algorithmically distinct enumeration routes) and by 351 other equivalent words (352 in total tie the maximum to machine precision); the gap to the Lüders bound is 2.4×10^{-4} . Random sampling at $L \geq 22$ yields $K_3 = 1.499964$, a gap of 3.6×10^{-5} , i.e. 99.998% of the Lüders bound. We never observe a violation of the Lüders bound, consistent with the structural impossibility for three projective measurements of a two-level system.

A characteristic nonmonotonicity in the best-at-exactly- L curve appears at $L = 5, 7, 10$, visible in Fig. 1(a). This is the expected fingerprint of a discrete dense group: the word set at length $L + 1$ is constructed afresh and need not contain the best length- L word, so the per-length maximum can decrease. No simple length-divisibility rule governs which L attains the saturating value: Table I shows $L = 9$ and $L = 11$ tied at the maximum $K_3 = 1.499762$, while $L = 6$, a central-element length $(\sigma_1\sigma_2)^{3k}$ with $k = 1$, reaches only $K_3 = 1.47744$, well below the $L = 9/11$ value (an earlier conjecture that lengths divisible by 6 are structurally favored is not supported by the data and is retracted here). The reverse-monotonic version “best at length $\leq L$ ” is monotone by construction.

B. Ising: exact $K_3 = 1$

Repeating the same search with the Ising generators gives a qualitatively different result: $K_3 = 1.000000$ for the best word at *every* $L \leq 11$ in the exhaustive search, and the same for every random sample at $L \in \{12, 14, \dots, 40\}$ (Fig. 1(a), red). The best Ising words realize $(C(B), C(B^2)) = (1, 1)$ for the best word at each $L \leq 10$ (the generators act as $\pm Z$ on the chosen basis), and $(0, -1)$ for the $L = 11$ argmax word (the gen-

erators map to $\pm X$), both of which give $K_3 = 1$ exactly.

This is a protocol-specific temporal counterpart of the Howard–Vala no-violation result [10] for the spatial CHSH inequality: in both cases the Ising-braiding subgroup of $SU(2)$ is too small to reach the bound. The intuition is the same: density in $SU(2)$ is, on the two-dimensional fusion space, necessary and sufficient for saturating the Lüders $3/2$ bound by braiding alone, just as it is necessary for saturating the Tsirelson $2\sqrt{2}$ bound by braiding alone; sufficiency at $d = 2$ rests on the numerical sweep of the companion study [19], not on an analytic proof. Ising braiding generates only Clifford gates and is missing a non-Clifford direction; the practical phase gate of Clarke, Sau, and Das Sarma [9] supplies exactly that missing resource in Majorana wires.

Cross-check against a search-space artifact. A clean $K_3 = 1.000000$ over the full tested range ($L \leq 11$ exhaustive, even $L \in \{12, \dots, 40\}$ random) raises the natural question whether the search has simply missed a violating word. To rule this out, we apply the same exhaustive procedure to the Ising generators deformed by $\delta \neq 0$ (an algebraic generalization in which the second R-symbol is rotated to $R_\psi e^{i\delta}$). The result is that the same code and word alphabet, searching all 4^9 words of length $L = 9$ at each of 49 grid points $\delta \in [0, 2\pi]$, recovers K_3 values across the full range $[1.000, 1.500]$ as δ varies, with $K_3 = 1.500$ reached at $\delta/\pi \approx 1.167$. A search that cannot find $K_3 > 1$ at $\delta = 0$ is therefore not search-limited; it is genuinely the standard Ising algebra that forbids the violation. (We do not interpret $\delta \neq 0$ Ising as a physical model; it is used here only as a falsification probe of the search procedure.)

We emphasize that the $K_3 = 1$ result is a statement about the present two-measurement-step protocol on σ -anyon braiding alone. A hybrid protocol that inserts an explicit non-Clifford resource into an Ising platform—or that uses an Ising-anyon implementation of a different gate set—could lift K_3 above 1. The result above is for pure braiding within the standard Ising fusion sector.

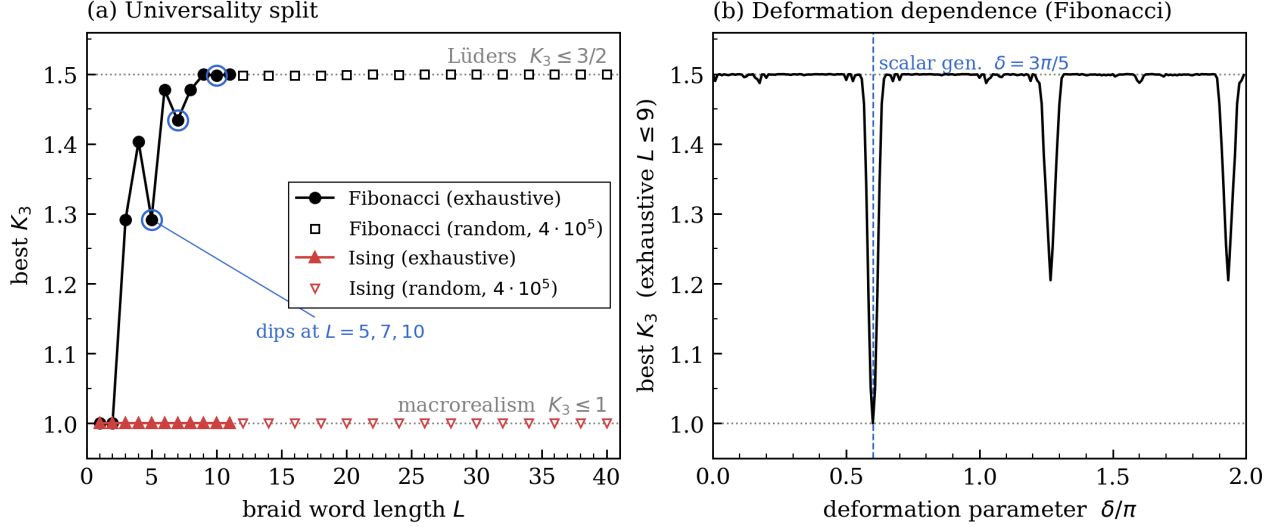


FIG. 1. (a) Best K_3 versus braid length L , $\delta = 0$. *Fibonacci* (black) saturates the Lüders bound $3/2$ to 99.998%; *Ising* (red) is pinned at 1.0 for every L tested (exhaustive to $L = 11$; random sampling, 4×10^5 words per L , at even $L \in \{12, 14, \dots, 40\}$). (b) Best $K_3(\delta)$ envelope for Fibonacci, exhaustive $L \leq 9$, fine sampling. The dashed line marks the algebraic singularity $\delta = 3\pi/5$ at which σ_1 collapses to a scalar (numerical scalar-deviation 3.1×10^{-16}). Data: `lgi_results.json` and `lgi_ising_results.json` for (a), `lgi_period_and_purestate_results.json` for (b); plotted by `lgi_letter_figure.py`.

C. The scalar singularity at $\delta = 3\pi/5$

The deformation parameter δ rotates R_τ along the unit circle. At the specific point $\delta_* = 3\pi/5$, the two diagonal entries of σ_1 coincide: $R_\tau e^{i\delta_*} = e^{3\pi i/5} e^{3\pi i/5} = e^{6\pi i/5} = R_1$, so σ_1 becomes a scalar times the identity. By $\sigma_2 = F\sigma_1 F$ and $F^2 = I$, σ_2 then collapses to the same scalar; every braid word is a global phase, and no $K_3 > 1$ is achievable for any L . We verify this numerically: $\|\sigma_1(\delta_*) - e^{i\theta} I\| = 3.1 \times 10^{-16}$. This is a clean, exact, machine-precision sub-result that follows from the algebra of the F- and R-symbols alone. This collapse threshold coincides with the intrinsic σ_1 rotation angle $\delta^*(k=3) = \pi k/(k+2) = 3\pi/5$ of the companion $SU(2)_k$ analysis [19]: both are the relative phase $\arg(R_1/R_\tau)$ of the two braid eigenvalues.

D. Dependence on the deformation parameter

Outside the singularity, exhaustive search with $L \leq 9$ explores the full δ axis (Fig. 1(b)). The range of best K_3 at fixed budget is $[1.0, 1.5]$, and the $K_3 = 1$ endpoint is attained only at the algebraic singularity $\delta^* = 3\pi/5$ itself: every nonsingular grid point violates the inequality already at $L \leq 9$, with the weakest envelope values ($K_3 \approx 1.051$) at the two grid points adjacent to the singularity ($\delta/\pi = 0.5917$ and 0.6083). Seeded random sampling at $\delta/\pi = 0.6083$ with longer words raises the best value found to $K_3 = 1.075$ at $L = 12$ and $K_3 = 1.148$ at $L = 24$ (`lgi_sector_budget_validator.py`), a monotone recovery consistent with a braid-budget de-

pression that deepens continuously toward the singular point; within the deposited budgets it remains well short of the Lüders bound there. The only point at which a violation is excluded exactly is the algebraic singularity above. The envelope is mirror-symmetric about the singular point, and hence also about $\delta = 8\pi/5$: $K_{3,\max}(\delta) = K_{3,\max}(6\pi/5 - \delta)$. The symmetry is exact: complex conjugation maps $\sigma_i(\delta)$, and hence every word, to its counterpart at $6\pi/5 - \delta$ up to a global phase, and the correlator (5) is invariant under $U \mapsto U^*$ because Z is real, so $K_3(\delta; w) = K_3(6\pi/5 - \delta; w)$ for every word w ; the deposited envelope satisfies it to within 7×10^{-15} , and word by word the deposited check finds at most 2.5×10^{-14} (`lgi_braid_relation_check.py`). In particular, $\delta = 6\pi/5$ reproduces the $\delta = 0$ value $K_3 = 1.499762$, and the two weakest grid points $\delta/\pi = 0.5917$ and 0.6083 are mirror images of each other.

Withdrawal of the Fourier-period analysis. Earlier versions of this letter reported a dominant Fourier period of the δ dependence: $k = 6$ in v1.0, computed over words of exact length $L = 9$ (it reproduces under that definition, contrary to what v1.1 and v1.2 stated), and $k = 3$ in v1.1 and v1.2 for the envelope $K_{3,\max}(\delta) = \max_{|w| \leq L} K_3(\delta; w)$, attributed there to reshuffling of the optimal word across the sweep. The envelope itself is not $2\pi/3$ -periodic: $K_{3,\max} = 1$ only at $\delta = 3\pi/5$, whereas $K_{3,\max}(3\pi/5 + 2\pi/3) = 1.2047$; $k = 3$ is only its largest Fourier component. An exhaustive audit of all 349,524 words with $|w| \leq 9$ on the same grid (`lgi_envelope_word_audit.py`) shows that this component is not specific to the maximization: no single word attains the envelope at every δ , but 4,420

of the 19,362 words that attain it at some grid point other than $\delta = 3\pi/5$ already have $k = 3$ as the strongest harmonic of their own curve $K_3(\delta; w)$. Most fixed-word curves are 2π -periodic, but not all: 42 of the 924 non-constant curves in a random sample of 1000 words have a shorter period. Since, in addition, the deformed generators do not form a braid-group representation at generic δ (Sec. II), we withdraw the period analysis and assign no physical meaning to the Fourier content of the envelope.

E. Initial-state independence (consistency check)

Equation (5) uses the maximally mixed initial state. As a consistency check on that formula, we have repeated K_3 for the optimal $L = 11$ word with pure initial states $|0\rangle$, $|1\rangle$, $|\pm\rangle$, $|\pm i\rangle$, and three Haar-random pure states. All nine give the same value to machine precision:

initial state	K_3	spread
$ 0\rangle, 1\rangle, \pm\rangle, \pm i\rangle$	1.499762	—
3 Haar-random pure $ \psi\rangle$	1.499762	—
mean over pure states	1.499762	$ \text{mean} - \text{mixed} = 0$

This is not a special property of the Lüders-saturating word: for two dichotomic (± 1 -eigenvalue) observables Q_1, Q_2 measured sequentially by projective (Lüders) measurements starting from an arbitrary state ρ , the two-time correlator is exactly $C(Q_1, Q_2) = \text{Tr}[\rho \frac{1}{2}\{Q_1, Q_2\}]$ in any dimension. On the two-dimensional fusion space used here, $Q_1 = Z$ and $Q_2 = UZU^\dagger$ are *both* traceless Hermitian operators with eigenvalues ± 1 for *every* unitary U (Bloch-vector operators on a qubit), and any two such operators have an anticommutator proportional to the identity, $\{Q_1, Q_2\} \propto I$ — a generic $d = 2$ Pauli-algebra identity, not a property singled out by the Lüders-saturating word. Consequently $C(U)$, and hence K_3 , is exactly independent of ρ for every braid word U , optimal or not. The measurement above is therefore a consistency check on Eq. (5) rather than an independent finding: it confirms that the deposited code correctly implements the trace formula, and pre-empts a natural reviewer concern that the mixed-state convention might mask a more nuanced state dependence.

IV. DISCUSSION

What the universality split says. The pair (Fibonacci, Ising) is the natural minimal test bed for universality-driven phenomena in non-Abelian anyon platforms. Howard and Vala [10] established that Ising braiding alone cannot violate the spatial CHSH inequality (any violation requires a non-stabilizer operation—gates and measurements together); Clarke, Sau, and Das Sarma [9] supply a practical phase gate providing exactly that missing resource in Majorana wires. The universal Fibonacci model, by contrast, saturates the bound by den-

sity [7], Brennen *et al.* [11] constructing explicit (sub-Tsirelson) CHSH-violating settings. The present results show that the same split holds for the *temporal* inequality: Fibonacci saturates the Lüders bound to 99.998% while Ising is pinned at 1.0 on the two-dimensional fusion space. In a single sentence: under projective Lüders measurement from a maximally mixed state on the two-dimensional fusion space, universality is a property that allows a non-Abelian anyon braid to saturate both the spatial and the temporal projective-quantum bounds.

What it does not say. The Fibonacci Lüders saturation is not, by itself, evidence for a nonclassical *topological* mechanism beyond what the modular tensor category already prescribes. Fibonacci density in $\text{SU}(2)$ guarantees that any unitary—including the one realizing the Lüders-saturating product—can be approximated by a braid word; the role of topology here is to provide that universal gate set, not to add an additional nonclassical resource. The Leggett–Garg saturation is a TQFT-consistency demonstration, in the same sense in which the spatial CHSH saturation is a TQFT-consistency demonstration. The interest of the result lies in (i) closing the non-Abelian Bell/Leggett–Garg literature gap, (ii) the sharpness of the Fibonacci/Ising split, and (iii) the algebraic scalar singularity at $\delta = 3\pi/5$. K_3 is computed here for projective Lüders measurement of Q at every time step and a maximally mixed (or, equivalently by Sec. III, any pure) initial state; its behavior under weak or nonprojective measurement protocols is not addressed and is left open.

Relation to Gómez-Ruiz et al. 2018. Gómez-Ruiz *et al.* [12] test the three-time LGI in the form $K_{i,j}(t) = 2C_{i,j}(t) - C_{i,j}(2t) \leq 1$ on Dirac qubits formed from edge Majorana modes of the Kitaev chain. That system is abelian; the test serves as a probe of the topological phase transition. Their maximum reported violation $K_{\text{Max}}(\mu)$ reaches ≈ 1.5 across a chemical-potential sweep but is not analyzed as a saturation distance to the Lüders bound. The present work tests K_3 on the non-Abelian fusion-channel qubit of three τ anyons; the Fibonacci-Ising universality split is the central comparison; and the saturation is reported as a 3.6×10^{-5} gap to the Lüders bound (i.e. 99.998%) obtained by braid-word optimization, rather than as a transition diagnostic. Both works belong to the genre “Leggett–Garg on a topological system”; they are methodologically and physically distinct sub-cases. Gómez-Ruiz *et al.* nonetheless serves as the relevant reference point for this line of work. Their setting is abelian, and braiding in such Majorana realizations is confined to the Clifford group (a property of the platform, not a claim made in their work); it is the one published precedent against which any non-Abelian, universal-braiding LGI test—including the companion $\text{SU}(2)_k$ study discussed below—must be distinguished.

What an experiment would look like. A direct test requires (i) a platform supporting non-Abelian braiding—present candidates are filling-fraction $\nu = 12/5$ for Fi-

bonacci [7] and quantum-processor implementations of Fibonacci string-nets [16, 17]; (ii) projective measurement of the fusion channel of an anyon pair [8]; (iii) the ability to apply a deterministic braid word between measurements. The Howard–Vala no-violation result predicts that for this Ising braid representation, under the present sequential B_3 Lüders protocol on the two-dimensional fusion space, the K_3 would be pinned at 1 even in the absence of decoherence; the digital topological realizations of Refs. [14, 15] implement a different, spatial setting. The robustness of K_3 to depolarizing noise on the platform benchmark of [14] would have to be analyzed separately, following the decoherence-threshold framework of [5]. An experimental test would additionally have to address the clumsiness loophole—that a violation could be caused by the measurements disturbing the system [3, 20].

V. OPEN QUESTIONS

Three natural extensions are immediate but outside the scope of this note:

- O1.** K_n for $n > 3$. For n equally spaced projective measurements the Lüders bound is $K_n^{\max} = n \cos(\pi/n)$ [3], approaching the algebraic limit $K_n \rightarrow n$ as $n \rightarrow \infty$. Whether Fibonacci braiding saturates this moving bound at each n is untested here; a multi-time Leggett–Garg test with $n = 4, 5, \dots$ is a one-day extension of the present setup.
- O2.** Structural origin of the nonmonotonicity dips. The proximate cause of the dips at $L = 5, 7, 10$ is geometric: at those lengths the best reachable braid word lands farther from the K_3 -optimal rotation than at neighboring lengths, so the per-length maximum drops. Whether this reachability pattern is governed at a deeper level by the divisibility structure of the central element $(\sigma_1\sigma_2)^3$ of B_3 is left open; a systematic correlation of dip-position with the algebra would be a sharper version of “best at length L ” than the running maximum.
- O3.** Representation dependence of the temporal bound. Sorella [13] has shown that in dimensions higher than two the Bell operators admit unitarily inequivalent representations, so that the *attained* CHSH violation is representation-dependent, with Tsirelson’s bound as the invariant maximum. The temporal counterpart—whether the Lüders bound $3/2$ is representation-dependent in $\dim > 2$ fusion spaces—is open. Our setup is fixed at $\dim = 2$; a $\dim > 2$ extension would address this directly. This question is supplemented by a companion work on the $SU(2)_k$ family across braid-representation dimension [19].

NOTATION

Symbols used in this paper.

Symbol	Meaning
L	braid-word length
σ_1, σ_2	generators (braid generators at $\delta = 0$), $\sigma_1 = \text{diag}(R_1, R_\tau e^{i\delta})$, $\sigma_2 = F \sigma_1 F$
F, R_1, R_τ	standard Fibonacci F - and R -symbols, $R_1 = e^{-4\pi i/5}$, $R_\tau = e^{+3\pi i/5}$
δ	algebraic phase-deformation parameter of the generators; $\delta = 0$ recovers the standard Fibonacci theory
$C(B), C(B^2)$	two-time correlators of Lüders measurements of $Q = \sigma_z$ separated by the braid B and by B^2
K_3	three-time Leggett–Garg combination, $K_3 = 2C(B) - C(B^2)$

DATA AND CODE AVAILABILITY

Paper, figures, and data are released under the Creative Commons Attribution 4.0 International (CC BY 4.0) license; the deposited source code is released under the Apache License 2.0 (LICENSE and LICENSE-CODE in the record; <https://creativecommons.org/licenses/by/4.0/>). All code and result files are deposited with this record [18]. Reproduction is deterministic with the random seeds 20260524 (Fibonacci main), 20260525 (Ising, period scan, and pure-state robustness), 20260723 (sector budget validator, `lgi_sector_budget_validator.py`), and 20260928 (envelope word audit, `lgi_envelope_word_audit.py`). The braid-relation and mirror-symmetry checks of Secs. II and IIID are reproduced by `lgi_braid_relation_check.py`. All computations use only `numpy` and `matplotlib`. Sanity checks at the head of `lgi_fibonacci.py` and `lgi_ising.py` (unitarity, Yang–Baxter, $(\sigma_1\sigma_2)^3$ scalar, all at $\delta = 0$) must pass at machine precision; if they do not, the generator conventions have been altered.

USE OF AI TOOLS

In the research, processing, and writing of this paper and its results I worked together with generative AI tools, in practice a system of multiple coordinated AI instances that I set up and orchestrate (large language models, mainly Claude, by Anthropic, inside Claude Code). At their current context sizes I found it far more effective to work with several specialized instances, each with its own role and its own harness of rules and parameters that I designed and refined through feedback, than to load a single instance with all of the material; for my workflow that would have been inefficient, though this depends

on the individual implementation. I lead this collaboration: I choose the research directions, set the goals, and make the final decisions in open exchange with the AI, learning actively as the work proceeds. The AI carries out the drafting, including the mathematical and technical parts, the numerical computation, and the literature search, under my direction. The AI works autonomously only task by task, within the structure I develop through feedback: it completes a task, and at open questions that need me it stops until the point is settled before the next step. Along the way I witness and take many of the decisions that shape the path, and it is common for me to spot things that need improvement. The work spans many separate runs, and a single simulation or build task alone can take up to an hour, so it could not happen all together in one autonomous run; and had I let the AI do all of it together alone, even if it is possible, it would no longer be my work but the AI's.

I run multiple verifications at the different stages of the

work and one before release, including cross-checks with an unrelated AI model from a different company, and all references are checked against the original sources. In the end what matters are human eyes, a principle that is itself written into the parameters of my system: I reach out to experts after publishing for review and feedback, so I learn what is solid and what must be corrected or falsified. My scripts for reproduction and review are released with this record. These tools are not authors; I am the author, and I take full responsibility for all scientific content and decisions leading to these results and their publication.

ACKNOWLEDGMENTS

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. The author declares no competing interests.

-
- [1] A. J. Leggett and A. Garg, *Quantum mechanics versus macroscopic realism: Is the flux there when nobody looks?* Phys. Rev. Lett. **54**, 857 (1985).
 - [2] C. Budroni and C. Emary, *Temporal quantum correlations and Leggett-Garg inequalities in multi-level systems*, Phys. Rev. Lett. **113**, 050401 (2014); [arXiv:1309.3678](#).
 - [3] C. Emary, N. Lambert, and F. Nori, *Leggett-Garg inequalities*, Rep. Prog. Phys. **77**, 016001 (2014); [arXiv:1304.5133](#).
 - [4] T. Fritz, *Quantum correlations in the temporal CHSH scenario*, New J. Phys. **12**, 083055 (2010); [arXiv:1005.3421](#).
 - [5] C. Emary, *Decoherence and maximal violations of the Leggett-Garg inequality*, Phys. Rev. A **87**, 032106 (2013); [arXiv:1212.3865](#).
 - [6] J. Kofler and Č. Brukner, *The conditions for quantum violation of macroscopic realism*, Phys. Rev. Lett. **101**, 090403 (2008); [arXiv:0706.0668](#).
 - [7] C. Nayak, S. H. Simon, A. Stern, M. Freedman, and S. Das Sarma, *Non-Abelian anyons and topological quantum computation*, Rev. Mod. Phys. **80**, 1083 (2008).
 - [8] P. Bonderson, M. Freedman, and C. Nayak, *Measurement-only topological quantum computation*, Phys. Rev. Lett. **101**, 010501 (2008).
 - [9] D. J. Clarke, J. D. Sau, and S. Das Sarma, *A practical phase gate for producing Bell violations in Majorana wires*, Phys. Rev. X **6**, 021005 (2016); [arXiv:1510.00007](#).
 - [10] M. Howard and J. Vala, *Nonlocality as a benchmark for universal quantum computation in Ising anyon topological quantum computers*, Phys. Rev. A **85**, 022304 (2012); [arXiv:1112.1516](#).
 - [11] G. K. Brennen, S. Iblisdir, J. K. Pachos, and J. K. Slingerland, *Non-locality of non-Abelian anyons*, New J. Phys. **11**, 103023 (2009); [arXiv:0810.4319](#).
 - [12] F. J. Gómez-Ruiz, J. J. Mendoza-Arenas, F. J. Rodríguez, C. Tejedor, and L. Quiroga, *Universal two-time correlations, out-of-time-ordered correlators and Leggett-Garg inequality violation by edge Majorana fermion qubits*, Phys. Rev. B **97**, 235134 (2018); [arXiv:1802.10522](#).
 - [13] S. P. Sorella, *On the representations of Bell's operators in Quantum Mechanics*, Found. Phys. **53**, 59 (2023); [arXiv:2304.05696](#).
 - [14] T. I. Andersen *et al.* (Google Quantum AI and Collaborators), *Non-Abelian braiding of graph vertices in a superconducting processor*, Nature **618**, 264 (2023); [arXiv:2210.10255](#).
 - [15] M. Iqbal, N. Tantivasadakarn, R. Verresen, S. L. Campbell, J. M. Dreiling, C. Figgatt, J. P. Gaebler, J. Johansen, M. Mills, S. A. Moses, J. M. Pino, A. Ransford, M. Rowe, P. Siegfried, R. P. Stutz, M. Foss-Feig, A. Vishwanath, and H. Dreyer, *Non-Abelian topological order and anyons on a trapped-ion processor*, Nature **626**, 505 (2024).
 - [16] S. Xu, Z.-Z. Sun, K. Wang, H. Li, Z. Zhu, H. Dong, J. Deng, X. Zhang, J. Chen, Y. Wu, C. Zhang, F. Jin, X. Zhu, Y. Gao, A. Zhang, N. Wang, Y. Zou, Z. Tan, F. Shen, J. Zhong, Z. Bao, W. Li, W. Jiang, L.-W. Yu, Z. Song, P. Zhang, L. Xiang, Q. Guo, Z. Wang, C. Song, H. Wang, and D.-L. Deng, *Non-Abelian braiding of Fibonacci anyons with a superconducting processor*, Nat. Phys. **20**, 1469 (2024); [arXiv:2404.00091](#).
 - [17] Z. K. Mineev, K. Najafi, S. Majumder, J. Wang, A. Stern, E.-A. Kim, C.-M. Jian, and G. Zhu, *Realizing string-net condensation: Fibonacci anyon braiding for universal gates and sampling chromatic polynomials*, Nat. Commun. **16**, 6225 (2025); [arXiv:2406.12820](#).
 - [18] B. Y. Sayim, *Code and data for: Leggett-Garg saturation and structural signatures in Fibonacci-anyon braiding*, Zenodo (2026); [doi:10.5281/zenodo.20372744](#).
 - [19] B. Y. Sayim, *A dichotomous Leggett-Garg witness for braid-representation density in $SU(2)_k$: exact at $d = 2$, dimension-limited at $d \geq 3$* , Zenodo preprint (2026); [doi:10.5281/zenodo.20531124](#).
 - [20] M. M. Wilde and A. Mizel, *Addressing the clumsiness loophole in a Leggett-Garg test of macrorealism*, Found. Phys. **42**, 256 (2012); [arXiv:1001.1777](#).